

2022-23

DISCRETE MATHEMATICS

CSC - 301

Full Marks : 60

Time : Three Hours

*The figures in the margin indicate full marks.*Attempt *all* the questions.

Write the answer of all the sub-parts
of a question together in order.

1. (a) Without using truth table, show that
 $(P \rightarrow (Q \rightarrow R)) \Leftrightarrow ((P \wedge Q) \rightarrow R)$.
- (b) Prove that $(\exists x)(P(x) \wedge Q(x)) \Leftrightarrow (\exists x)P(x) \wedge (\exists x)Q(x)$. Also justify whether the following is true or not?

$$(\exists x)P(x) \wedge (\exists x)Q(x) \Leftrightarrow (\exists x)(P(x) \wedge Q(x))$$

- (c) An island has two tribes of natives. Any native from the first tribe always tells the truth, while any native from the other tribe always lies. You arrive at the island and ask a native if there is gold on the island. He answers, "There is gold on the island if and only if I always tell the truth." Which tribe is he from? Is there gold on the island?

$$4 + (2 + 2) + 2 = 10 \text{ [CO1, CO2]}$$

P.T.O.

-(2)

2. How many points (i.e. minimum number of points) must be chosen inside an equilateral triangle of side-length 3 unit in order to ensure that at least one pair are not more than distance 1 unit apart? Explain your answer. 3 [CO3, CO4]

Or

Let $m, n \in \mathbb{N}$. If m balls are placed in n boxes, then prove that there must exist a box containing at least $\left\lceil \frac{m}{n} \right\rceil$ balls.

3. You walk into a candy store and have enough money for 6 pieces of candy. The store has chocolate (C), gummies (G) and horrible Chinese candy (H). How many different selections can you make? 4 [CO1, CO2]

Or

A string of parentheses is valid if there are an equal number of open and closed parentheses and if you begin at the left as you move to the right, add 1 each time you pass an open and subtract 1 each time you pass a closed parenthesis, then the sum is always non-negative. For example, “(())()” is valid, but “())()” is not. How many groupings are there for each value of n ? Compute this value with explanation.

4. In a town there are three newly build houses and each needs to be connected by a line to the gas, water and electricity factories. The lines are only allowed to run along the ground. Is there a way to make all nine connections without any of the lines crossing each other? You must explain your answer. 4 [CO2, CO3]

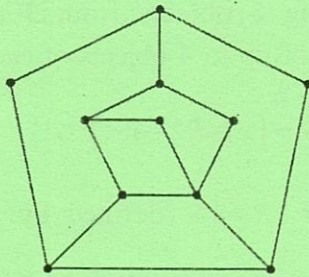
(3)

5. n couples arrived at a party and were greeted by the host and the hostess at the door. After several rounds of handshaking the host asked the guests as well as the hostess (his wife) to indicate the number of hands each one of them had shaken. He got different answers from each one. Given that no one shook hands with his or her own spouse, how many hands had the hostess shaken? Explain your answer.

4 [CO2, CO3, CO4]

Or

Show that the following graph has no hamiltonian circuit.



6. Draw the Hasse diagram of $(\{3, 4, 5, 60, 120, 240, 360, 720\}, |)$. Find the least element, greatest element, maximal element, minimal element, lower bound, upper bounds, glb and lub of the following sets :

(a) $A_1 = \{3, 4, 5\}$, (b) $A_2 = \{60, 120\}$ and (c) $A_3 = \{240, 360\}$.

10 [CO1, CO3, CO5]

7. (a) Show that the relation 'congruence modulo m ($\equiv$)' over the set of $\mathbb{Z}^+$ is an equivalence relation. Also show that if $a \equiv b$ and $c \equiv d$ then $(a+c) \equiv (b+d)$.

P.T.O.

(4)

- (b) Suppose X , Y and Z be any three sets. R is a relation from X to Y and S is a relation from Y to Z . Then prove that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$. 5+5 [CO2, CO3, CO4]

Or

- (a) Prove that 'Congruence modulo 4' is an equivalence relation defined on $\mathbb{Z}^+$.
- (b) If R and S are equivalence relations, then show that $(R \cap S)$ is also an equivalence relation.
8. (a) Using the generating function, find the number of ways of selecting 6 objects from 3 types of objects if repetitions of up to 4 objects of each type are allowed.
- (b) Suppose $S = \{x \in \mathbb{R} : -1 < x \leq 1\}$, then show that the function $f : \mathbb{R} \rightarrow S$ defined by $f(x) = \frac{x}{1+|x|}$ is a bijective function. 5+5 [CO2, CO3, CO4, CO5]

Or

- (a) Solve the recurrence relation $a_n = 7a_{n-1} - 13a_{n-2} - 3a_{n-3} + 18a_{n-4}$, where $a_0 = 5$, $a_1 = 3$, $a_2 = 6$, $a_3 = -21$.
- (b) A survey of 260 television viewers produced the following information : 64 watch sports, 58 watch films, 94 watch news, 28 watches both sports and films, 26 watches both sports and news items, 22 watches both films and news items and 14 are interested in all three types. How many

(5)

people in the survey are not interested in any of these three types? How many people in the survey are interested in seeing only news items?

Course Outcomes :

- CO1 : Remember the basic terms, definitions and concepts of mathematics.
- CO2 : Students will be able to understand the key concepts of discrete mathematics such as functional mapping, mathematical logic, counting principles, generating functions, algebraic structures and graph theory.
- CO3 : Students will be able to apply the learned concepts to solve various problems.
- CO4 : Students will be able to differentiate or relate the various ideas with respect to problems.
- CO5 : Students will be able to judge the formulas and ideas to be applicable to a problem.